

Responsible use of large language models for research: Literature research

From general understanding to a practical literature-search workflow

Lodewijk Pet	Clinical Epidemiology	PhD candidate	l.a.pet@lumc.nl
Bram van Dijk	Public Health and Primary Care	Postdoc	b.m.a.van_dijk@lumc.nl

Course website: <https://bma-vandijk.github.io/RULLMR/>

Programme

1. Recap of LLM fundamentals relevant to literature search
2. Intro of today's topic: literature search
3. Reflecting on and creating workflows
4. Designing and executing prompts for Copilot/ChatGPT/Gemini (deep research)
5. SLMs: setting up local tasks (homework assignment)
6. Tuning your local model for summarization

Learning goals

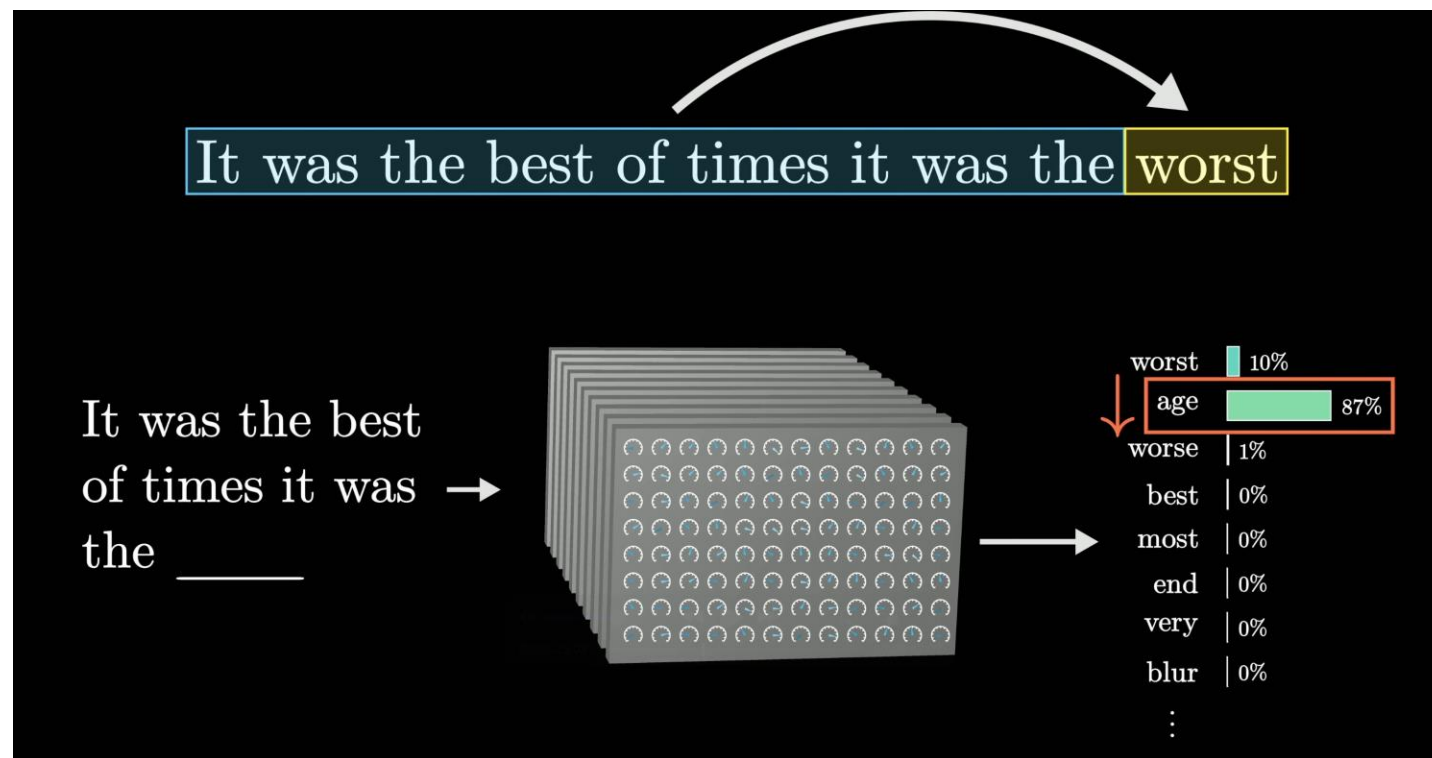
After this session you should be able to:

- Explain LLM fundamentals important to searching literature
- Explain where LLMs are and are not useful for searching literature
- Design a 'plan-first' workflow
- Design a prompt and do a (deep) search with (commercial) LLMs
- Set up a local model for basic literature work and identify limitations
- Explicate good research practices for responsible LLM-supported literature work

1. RECAP: LLM FUNDAMENTALS

1.1 LLMs predict next tokens

Remember:



www.3blue1brown.com

- An LLM predicts the next word
- Unlike humans, it does not necessarily distinguish between statistical likelihood and truth (though SOTA research shows some awareness of confidence) (e.g. Kumaran et al. 2026)
- It is not necessarily optimised for scientific validity

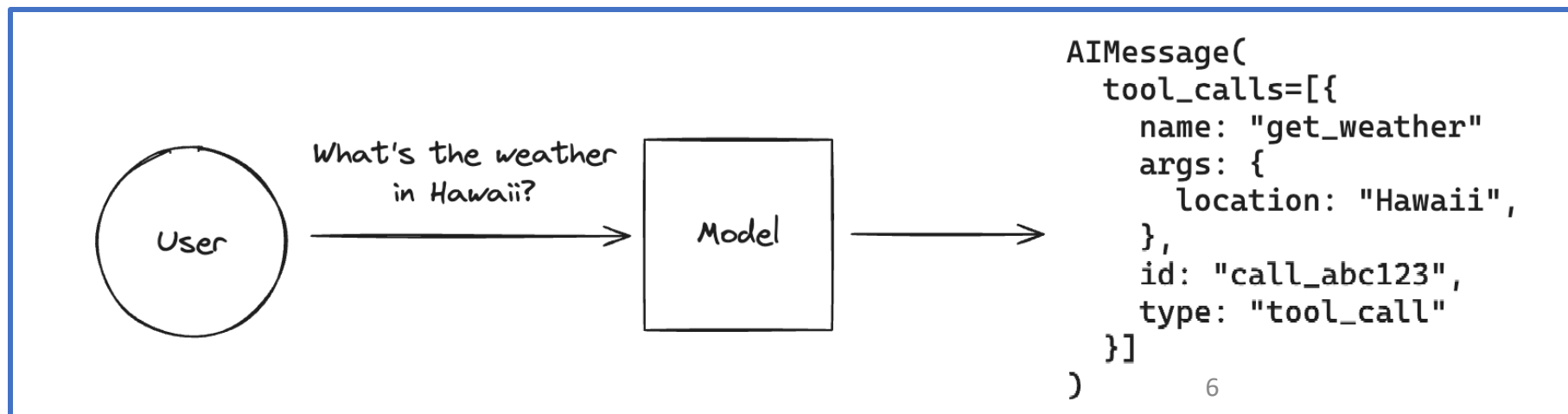
1.2 But not just natural language

LLM on its own

- Cannot do web search, code execution, file handling, or reference validation (though it *can* return outputs calling for function execution)
- Can therefore sound plausible without having access to current evidence
- Easy to run locally (smaller models)

LLM in harness/environment (agent)

- Harness/environment provides:
 - Runtime that acts on LLM outputs
 - Specification of available tools
 - Sub-agents (sometimes)
 - Extensive memory (e.g. tool call results)
- Copilot/ChatGPT/Claude are not just LLMs
- More complex to run, hence often cloud-based

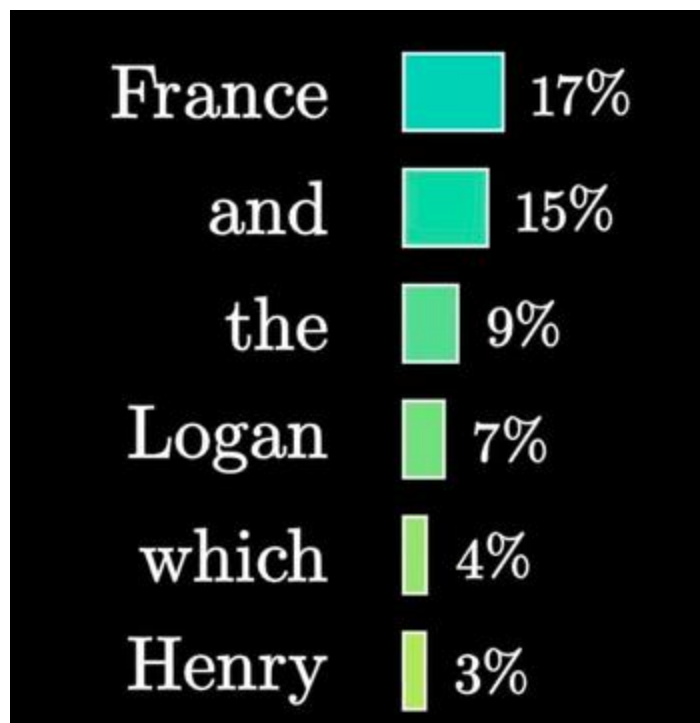


1.3 What do you think?

So what would be the best choice for literature search and why?

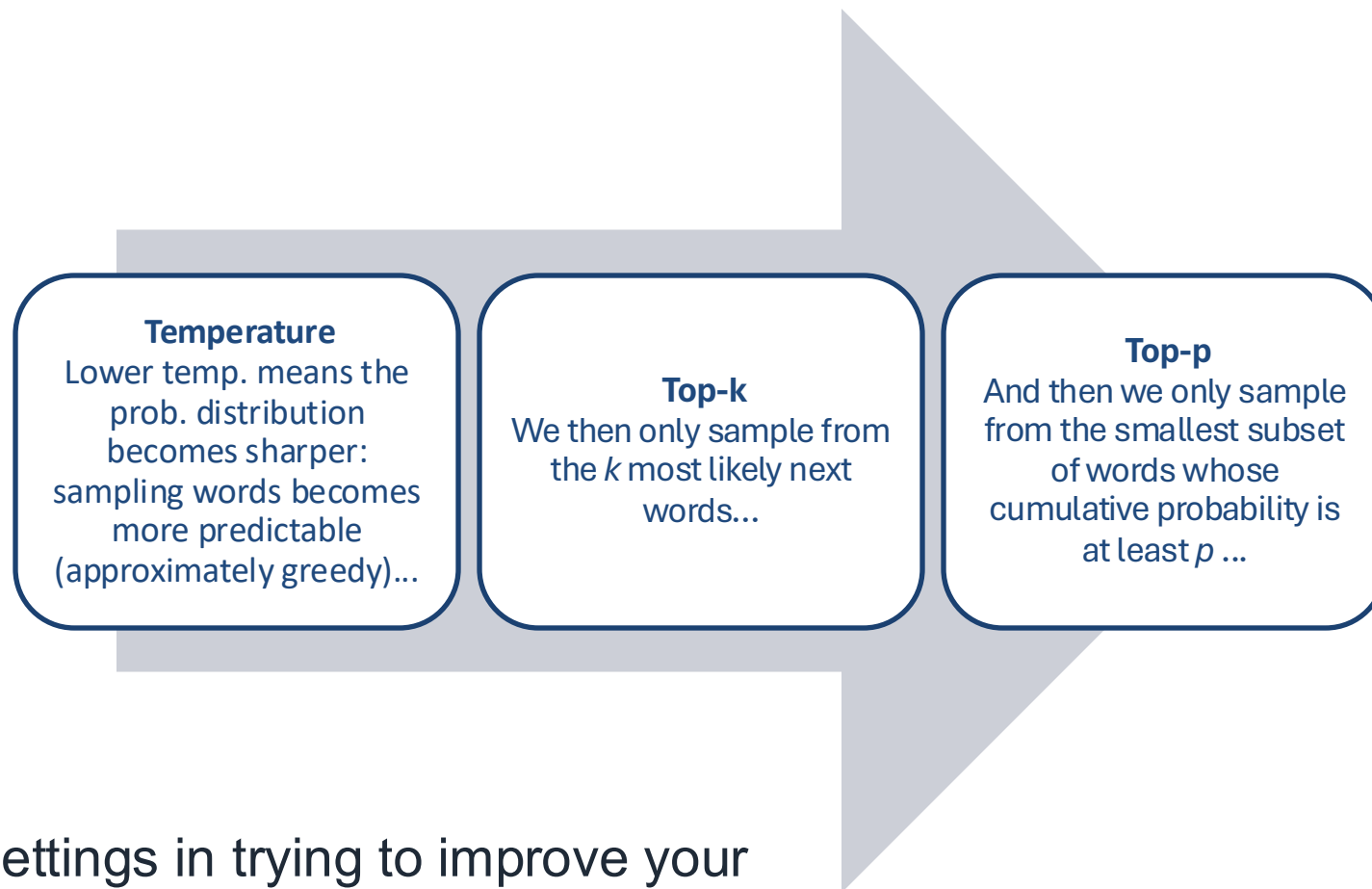
1.4 Sampling vs. model behaviour

- LLM as neural networks yield (virtually) always the same probability scores for the same prompts; **sampling** from these scores causes variation in output



www.3blue1brown.com

- You will experiment with these settings in trying to improve your LLM-generated summary



1.5 Hot off the press: LUMC AI policy for supportive us

- New policy covers only **supportive** use of GenAI
 - Summarizing, translating, drafting, ideation, not research, diagnostics, triage, prediction, etc.
 - Policy on GenAI in **research** is work in progress
- More possibilities with sensitive data and Copilot
 - E.g. internal or patient data
 - E.g. simplifying a letter for a patient or summarizing an important meeting
- Use of other tools is not prohibited, but a good evaluation of risk and balance needs to be made
 - E.g. create a TODO list for a low-stake meeting on education or research practicals
- **Look it up on Zenya and study it** (we also have a provisional translation)
(<https://lumc.zenya.work/QC/MM-286-X>)



2. TOPIC: LITERATURE SEARCH

2.1 Literature research as useful use case for LLMs

- Searching literature is knowledge-heavy, iterative, and repetitive.
- Pre-agentic LLMs already had excellent semantic skills
- Current LLMs in agentic harnesses excel in planning, searching, summarizing
- Much of the relevant information is public, especially at abstract level

2.2 Literature search vs Systematic Literature Review

- We are focussing on a more exploratory, 'informal' literature search for identifying research gaps, writing an introduction for a research paper, getting a quick overview for a specific topic
- Full SLR less suitable with LLMs
 - Why?

2.2 Literature search vs Systematic Literature Review

- We are focussing on a more exploratory, 'informal' literature search for identifying research gaps, writing an introduction for a research paper, getting a quick overview for a specific topic
- Full SLR less suitable with LLMs:
 - Process involves many steps that currently require **human justification** (scope of query, choice of databases, inclusion/exclusion) and **reproducibility** (hard with commercial models)
 - Some steps could be supported by LLMs (like expanding your query, identifying databases)

2.3 Question for you

- How do you currently go about searching literature for your introduction when writing a research paper (without using AI)?

2.4 Question for you

- How do you currently go about searching literature for your introduction when writing a research paper (without using AI)?
 - In AI, at least, it is often a mix of domain knowledge and convenience-based searching on Scholar, so results are cherry-picked
- LLMs can provide a broader search (but have their own biases) (Rao, Wong, Callison-Burch 2026) [1]

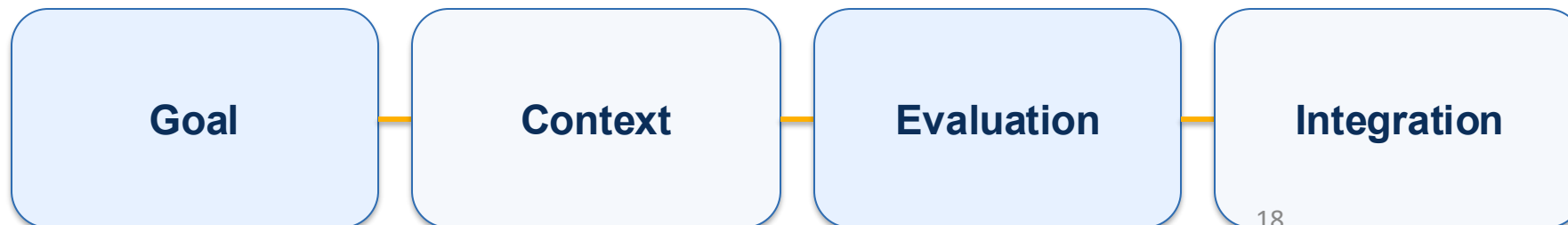
3. REFLECTING ON WORKFLOWS

3.1 For now, choose a research topic you know well

- Use a topic where you can judge whether the output is interesting or wrong.
- A good choice can be the title of a recent paper from your own field.
- Avoid starting with a topic where you cannot recognize a bad answer.
- Your own knowledge is part of the prompt context.

3.2 Answer the following questions about your topic

- Instead of directly prompting a literature search on your chosen topic, first ask...
 - What is my goal and what do I want to log? (goal)
 - What kind of literature do I need? (goal)
 - What is important for me to verify manually? (goal)
 - What are my criteria for success? (goal)
 - What do I already know about this topic? (context)
 - What could be useful tools? (context)
 - What is the quality of the output? (evaluation)
 - What will I use the final output for? (integration)



3.3 Using LLMs for specific parts of the workflow

Follow-up prompt

Search for recent chatbot features that can do a deep search for academic literature.

For each feature, provide:

- 1) what it does
- 2) what access is needed
- 3) limitations and failure modes
- 4) what I should verify manually.

Be concise.

What to look for

A static model answer may already be outdated.

Make current tool use explicit.

Ask about limitations, not just features.

Use this step to learn the tool landscape.

3.4 Choose a tool

- You can always choose LUMC's Copilot if you don't like to use personal accounts for access to Gemini/ChatGPT deep research modes
- Nice if at least some people use Copilot so that we can see the difference

4. PRACTICUM: DESIGNING PROMPTS

4.1 Practical steps



4.2 - Let the LLM design a prompt

Prompt for generating a prompt

I want you to help me use Gemini Deep Research to do a proper literature search for writing an introduction of a biomedical article.

Topic: "Training word2vec models from clinical text"

Give me an elaborate prompt that I can paste directly into the research mode. Force transparent steps, explicit source listing, uncertainty flags, no invented citations, and a final section called "What to verify manually".

What to look for

- Using the LLM to prompt is often faster than prompting from scratch.
 - LLMs can extrapolate best practices from training data, e.g. formatting, order.
- Read the generated prompt critically before running it. (It may be eager to do many things)
- Note: the same resulting prompt can also be used in 'normal' mode

4.3 - Execute your prompt

Resulting generated prompt

You are conducting a systematic, transparent literature-mapping exercise to support the Introduction section of a high-impact biomedical informatics manuscript.

TOPIC

"Training word2vec models from clinical text"

OBJECTIVE

Produce a comprehensive literature review focused on methods, datasets, validation approaches, applications, advantages, limitations, and historical evolution of training Word2Vec models from clinical text corpora.

This is NOT a narrative review written for publication. This is an evidence-gathering and field-mapping exercise that will be used later to write an Introduction.

=====
CORE REQUIREMENTS
=====

[...]

What to look for

- Carefully check the prompt; you can often only use Deep Research mode one or two times a day
- This is another reason why you need to think about your workflow before you start

4.4 - Understand research-capable modes

- They search, gather, and synthesize rather than only chat.
- They are highly relevant for public-source literature scouting.
- They can still miss papers, over-summarize, or stop too early.
- They should be judged by transparency, source quality, and fit to the task.

4.5 - Inspect and discuss results with your partner

- Check whether the report is transparent about steps and uncertainty.
- See whether the sources are explicit and relevant.
- Notice whether the model narrows the field too quickly.
- Ask whether the output matches your actual writing goal.

4.6 - Inspect and discuss results with your partner

- Check whether the cited papers really exist.
- Check whether the source actually supports the claim.
- Check whether obvious key papers were missed.
- Check whether the output is useful for your Introduction rather than merely interesting.

4.7 Literature is public in part, closed in part

Publicly accessible in many cases

- Abstracts are usually available.
- Public guidelines and many reviews are accessible.
- Open-access papers are increasingly common.

Still constrained

- Many full papers remain behind paywalls.
- You cannot assume the model has lawful access to full text.
- This limits what a general cloud model can work with directly.

4.8 What cloud LLMs are very good at

- Broad topic exploration and terminology expansion.
- Turning rough ideas into structured search plans.
- Comparing concepts, study designs, and narrative options.
- Scouting recent public developments when browsing is enabled.
- Drafting prompts and intermediate summaries.

4.9 What local SLMs are very good at

- Keeping data local when privacy or governance matters.
- Handling short bounded extraction tasks.
- Allowing direct control over parameters and workflow.
- Supporting repeatable local experiments.

5. SLMS: SETTING UP LOCAL TASKS (HOMEWORK ASSIGNMENT)

5.1 Ollama homework

Simple local test

```
ollama pull qwen2.5:7b-instruct
ollama run qwen2.5:7b-instruct "Explain confounding
in exactly 40 words."
```

Then try a short bounded literature task:

- Summarize this Introduction.

Or if you've already done that, different section,
different question:

- What is the main finding?
- Extract the study design.

What to look for

- **For those who didn't have time to do the homework: try to do it now**
- **For those that did it: compare results with your teammate and exchange best practices**
- The goal is not frontier-level performance.
- Notice speed, clarity, and failure patterns.
- Use short text first, not full papers.

5.2 What local SLMs struggle with

- Full papers mean long contexts.
- Small models degrade quickly when instructions and contexts are long (context rot).
- Slow local processing can make broad literature work unrealistic.
- Incoherence and omission can be bigger problems than hallucination alone.

5.3 When a small local model is still worth trying

- When you need one specific piece of information from many texts.
- When the extraction task is clearly defined in advance.
- When privacy makes cloud use unacceptable.
- When rough triage is acceptable and must be followed by manual review.

6. TUNING YOUR LOCAL MODEL FOR SUMMARIZATION

6.1 Settings change behavior

- You can play around with the following settings to see how it affects the response based on your instruction
 - Context window: too small means important text falls out.
 - Temperature: lower values reduce variation and may help extraction.
 - Top-k and top-p: restricting token choice can improve behavior.
 - Bad output can be a settings problem, not just a model problem.
- Experiment with this interface to play with model settings in a more intuitive way
 - <https://bma-vandijk.github.io/RULLMR/schedule/>
 - Download zip
 - Extract in folder
 - Follow instructions in start_here.txt

6. Try to download and run this graphical UI

 **LLM sampling** LOCAL LAB

See what fits. Change how your model thinks.

Runs on your computer

Try an example

Ollama connected
4 local models available · http://127.0.0.1:11434

Interface

Ollama

Model

qwen3:4b

Server address

01 / YOUR INPUT

Source text

The material you want the model to work with.

1

Paste your text here...

0 tokens

Clear

Prompt

The instruction to apply to your source text.

2

For example: Summarize this text in three bullet points.

0 tokens

Clear

Text → Prompt

☒ Rainbow tokens

02 / MODEL SETTINGS

Make room for your ideas.

Reset defaults

Explore failure modes

Each preset starts from standard settings.

Tiny context
64 tokens · drops early input

High temperature
2.0 · wide-open sampling

Short output
8 tokens · cuts replies short

High repeat penalty
4.0 · forces unusual wording

Choose a preset to experiment with an extreme setting.

Context window

tokens

4096

Model capacity: 262,144 tokens
0 excludes all pasted input.

Maximum output

tokens

512

Generation limit; output also uses context.

Temperature

0.7

0 = focused · 2 = more varied

When context fills

Allow shifts · projected view

Stop mode preserves the complete sent input.

Top P

0.9

Probability mass to sample.

Top K

40

Candidate limit; 0 disables.

Min P

0

Relative probability cutoff.

Repeat penalty

1.1

1 = no penalty · high values distort wording.

Seed

-1

-1 = random

 Fresh context for every run

Please indicate presence



Thank you

Questions, discussion, and workflow examples are welcome.

Please keep testing with your own topic and compare outputs critically.

Contact: l.a.pet@lumc.nl